Taipei Veterans General Hospital

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Sample Information

Patient Name: 廖美萩
Gender: Female
ID No.: E201721621
History No.: 46238157
Age: 63

Ordering Doctor: DOC3054C 陳奕廷
Ordering REQ.: 0CBMXVK
Signing in Date: 2022/10/27

Path No.: S111-97954
MP No.: F22113
Assay: Oncomine Focus Assay
Sample Type: FFPE
Block No.: S110-18079A+B
Percentage of tumor cells: 30%

Reporting Doctor: DOC5466K 葉奕成 (Phone: 8#5466)

Note:

Sample Cancer Type: Non-Small Cell Lung Cancer

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Report Highlights

2 Relevant Biomarkers
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Relevant Non-Small Cell Lung Cancer Variants

Gene	Finding	Gene	Finding
ALK	None detected	NTRK1	None detected
BRAF	None detected	NTRK2	None detected
EGFR	***EGFR exon 20 insertion, EGFR c.2284-1G>C***	NTRK3	None detected
ERBB2	None detected	RET	None detected
KRAS	None detected	ROS1	None detected
MET	None detected		

Disclaimer: The data presented here is from a curated knowledgebase of publicly available information, but may not be exhaustive. The data version is 2022.09(005).The content of this report has not been evaluated or approved by FDA, EMA or other regulatory agencies.

Relevant Biomarkers

Tier	Genomic Alteration	Relevant Therapies (In this cancer type)	Relevant Therapies (In other cancer type)	Clinical Trials
IA	*EGFR exon 20 insertion* epidermal growth factor receptor Allele Frequency: 4.38%	**amivantamab** [1, 2] **mobocertinib** [1]	None	7
IA	*EGFR c.2284-1G>C* epidermal growth factor receptor Allele Frequency: 4.97%	None	None	3

Public data sources included in relevant therapies: FDA[1], NCCN, **EMA**[2], ESMO
Tier Reference: Li et al. *Standards and Guidelines for the Interpretation and Reporting of Sequence Variants in Cancer: A Joint Consensus Recommendation of the Association for Molecular Pathology, American Society of Clinical Oncology, and College of American Pathologists.* J Mol Diagn. 2017 Jan;19(1):4-23.
*** Includes biosimilars**

Alerts informed by public data sources: Contraindicated, Resistance

EGFR exon 20 insertion	**gefitinib*** [2] afatinib, dacomitinib, erlotinib, gefitinib

Public data sources included in alerts: FDA[1], NCCN, **EMA**[2], ESMO

Prevalent cancer biomarkers without relevant evidence based on included data sources

ERBB3 p.(D297N) c.889G>A, JAK3 p.(S493N) c.1478G>A, MYC p.(P74L) c.221C>T

Variant Details

DNA Sequence Variants

Gene	Amino Acid Change	Coding	Variant ID	Locus	Allele Frequency	Transcript	Variant Effect	Coverage
EGFR	p.(?)	c.2284-1G>C	.	chr7:55248985	4.97%	NM_005228.5	unknown	161
EGFR	p.(A767_V769dup)	c.2308_2309insCCAGCGTGG	COSM12376	chr7:55248998	4.38%	NM_005228.5	nonframeshift Insertion	160
MYC	p.(P74L)	c.221C>T	COSM1166664	chr8:128750684	43.37%	NM_002467.6	missense	883
ERBB3	p.(D297N)	c.889G>A	COSM941489	chr12:56482341	73.52%	NM_001982.4	missense	608
JAK3	p.(S493N)	c.1478G>A	.	chr19:17949163	16.82%	NM_000215.4	missense	1855
MTOR	p.(H1803=)	c.5409C>T	.	chr1:11190790	10.20%	NM_004958.4	synonymous	843
MTOR	p.(E1485*)	c.4453G>T	.	chr1:11217225	25.19%	NM_004958.4	nonsense	135
MTOR	p.(A1462V)	c.4385C>T	.	chr1:11217293	21.43%	NM_004958.4	missense	140
MYCN	p.(?)	c.-1323A>G	.	chr2:16080864	7.27%	NM_005378.6	unknown	55
ALK	p.(*1621R)	c.4861T>C	.	chr2:29416092	9.59%	NM_004304.5	stoploss	73
ALK	p.(T1516=)	c.4548C>T	.	chr2:29416405	7.75%	NM_004304.5	synonymous	129
ALK	p.(A1252S)	c.3754G>T	.	chr2:29432734	5.76%	NM_004304.5	missense	243
ALK	p.(V1147M)	c.3439G>A	.	chr2:29445394	40.00%	NM_004304.5	missense	625
ALK	p.(L1145=)	c.3433C>T	.	chr2:29445400	6.43%	NM_004304.5	synonymous	622
ALK	p.(G1125=)	c.3375C>A	.	chr2:29445458	7.64%	NM_004304.5	synonymous	641
RAF1	p.(H434=)	c.1302T>C	.	chr3:12632365	4.97%	NM_002880.3	synonymous	1007

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Variant Details (continued)

DNA Sequence Variants (continued)

Gene	Amino Acid Change	Coding	Variant ID	Locus	Allele Frequency	Transcript	Variant Effect	Coverage
RAF1	p.(R256K)	c.767G>A	.	chr3:12645702	21.15%	NM_002880.3	missense	1891
PIK3CA	p.(F82=)	c.246T>C	.	chr3:178916859	11.73%	NM_006218.4	synonymous	162
PIK3CA	p.(T1031A)	c.3091A>G	.	chr3:178952036	8.37%	NM_006218.4	missense	406
PDGFRA	p.(G313=)	c.939T>G	.	chr4:55133726	47.72%	NM_006206.6	synonymous	526
KIT	p.(T544=)	c.1632C>T	.	chr4:55593475	17.47%	NM_000222.3	synonymous	521
KIT	p.(A629V)	c.1886C>T	.	chr4:55594183	24.19%	NM_000222.3	missense	62
KIT	p.(L656=)	c.1968A>G	.	chr4:55594265	7.14%	NM_000222.3	synonymous	70
KIT	p.(L813=)	c.2439A>T	.	chr4:55599313	4.90%	NM_000222.3	synonymous	102
FGFR4	p.(P136L)	c.407C>T	.	chr5:176517797	99.32%	NM_213647.3	missense	296
FGFR4	p.(R416=)	c.1248A>G	.	chr5:176520329	28.74%	NM_213647.3	synonymous	87
FGFR4	p.(R466K)	c.1397G>A	.	chr5:176520552	23.41%	NM_213647.3	missense	786
ROS1	p.(G2014E)	c.6041G>A	.	chr6:117638400	8.28%	NM_002944.2	missense	314
ESR1	p.(V534A)	c.1601T>C	.	chr6:152419914	13.55%	NM_001122740.1	missense	812
MET	p.(Y84H)	c.250T>C	.	chr7:116339388	4.92%	NM_001127500.3	missense	244
MET	p.(L833=)	c.2499G>A	.	chr7:116403184	29.80%	NM_001127500.3	synonymous	1510
MET	p.(T1259A)	c.3775A>G	.	chr7:116423446	5.41%	NM_001127500.3	missense	296
SMO	p.(R302K)	c.905G>A	.	chr7:128845608	16.98%	NM_005631.5	missense	53
SMO	p.(G416S)	c.1246G>A	.	chr7:128846410	28.93%	NM_005631.5	missense	401
SMO	p.(T528=)	c.1584T>A	.	chr7:128850321	19.68%	NM_005631.5	synonymous	127
SMO	p.(T534A)	c.1600A>G	.	chr7:128850337	36.29%	NM_005631.5	missense	124
BRAF	p.(I551T)	c.1652T>C	.	chr7:140476754	9.27%	NM_004333.6	missense	205
BRAF	p.(A308T)	c.922G>A	.	chr7:140500220	99.71%	NM_004333.6	missense	340
BRAF	p.(S273=)	c.819T>C	.	chr7:140501253	9.39%	NM_004333.6	synonymous	213
BRAF	p.(R271C)	c.811C>T	.	chr7:140501261	68.69%	NM_004333.6	missense	214
BRAF	p.(D213N)	c.637G>A	.	chr7:140507834	11.87%	NM_004333.6	missense	1087
MYC	p.(S129G)	c.385A>G	.	chr8:128750848	7.63%	NM_002467.6	missense	485
MYC	p.(Q380=)	c.1140G>A	.	chr8:128752979	97.56%	NM_002467.6	synonymous	409
GNAQ	p.(G207R)	c.619G>A	.	chr9:80409495	74.17%	NM_002072.5	missense	271
GNAQ	p.(R166H)	c.497G>A	.	chr9:80412544	5.04%	NM_002072.5	missense	139
FGFR2	p.(L376=)	c.1128G>A	.	chr10:123274790	15.91%	NM_000141.5	synonymous	729
HRAS	p.(Q25R)	c.74A>G	.	chr11:534249	29.86%	NM_001130442.2	missense	144
CCND1	p.(A121Pfs*25)	c.360_361delGGinsA	.	chr11:69457960	7.88%	NM_053056.3	frameshift Block Substitution	203

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Variant Details (continued)

DNA Sequence Variants (continued)

Gene	Amino Acid Change	Coding	Variant ID	Locus	Allele Frequency	Transcript	Variant Effect	Coverage
ERBB3	p.(A90=)	c.270C>T	.	chr12:56478814	34.53%	NM_001982.4	synonymous	921
NF1	p.(D973Y)	c.2917G>T	.	chr17:29556919	9.60%	NM_001042492.3	missense	677
ERBB2	p.(T686M)	c.2057C>T	.	chr17:37879682	79.33%	NM_004448.3	missense	421
ERBB2	p.(L1017=)	c.3051G>A	.	chr17:37883148	61.36%	NM_004448.3	synonymous	88
ERBB2	p.(S1049N)	c.3146G>A	.	chr17:37883243	35.40%	NM_004448.3	missense	113
BRCA1	p.(R7C)	c.19C>T	.	chr17:41276095	4.85%	NM_007294.4	missense	412
BRCA1	p.(L6V)	c.16C>G	.	chr17:41276098	4.85%	NM_007294.4	missense	412
BRCA1	p.(M1?)	c.3G>A	.	chr17:41276111	4.67%	NM_007294.4	missense	407
BRCA1	p.(?)	c.-1A>G	.	chr17:41276114	4.69%	NM_007294.4	unknown	405
MAP2K2	p.(E45=)	c.135A>G	.	chr19:4117585	5.37%	NM_030662.4	synonymous	354
MAP2K2	p.(L44=)	c.132G>A	.	chr19:4117588	5.08%	NM_030662.4	synonymous	354
JAK3	p.(A650=)	c.1950C>T	.	chr19:17945989	5.02%	NM_000215.4	synonymous	1694
JAK3	p.(K556=)	c.1668G>A	.	chr19:17948774	86.84%	NM_000215.4	synonymous	395
AR	p.(R789=)	c.2367G>A	.	chrX:66941723	9.15%	NM_000044.6	synonymous	2000

Biomarker Descriptions

EGFR (epidermal growth factor receptor)

Background: The EGFR gene encodes the epidermal growth factor receptor (EGFR) tyrosine kinase, a member of the ERBB/human epidermal growth factor receptor (HER) family. In addition to EGFR/ERBB1/HER1, other members of the ERBB/HER family include ERBB2/HER2, ERBB3/HER3, and ERBB4/HER4[1]. EGFR ligand induced dimerization results in kinase activation and leads to stimulation of oncogenic signaling pathways including the PI3K/AKT/MTOR and RAS/RAF/MEK/ERK pathways. Activation of these pathways promote cell proliferation, differentiation, and survival[2,3].

Alterations and prevalence: Recurrent somatic mutations in the tyrosine kinase domain (TKD) of EGFR are observed in approximately 10-20% of lung adenocarcinoma, and at higher frequencies in never-smoker, female, and Asian populations[4,5,6,7]. The most common mutations occur near the ATP-binding pocket of the TKD and include short in-frame deletions in exon 19 (EGFR exon 19 deletion) and the L858R amino acid substitution in exon 21[8]. These mutations constitutively activate EGFR resulting in downstream signaling, and represent 80% of the EGFR mutations observed in lung cancer. A second group of less prevalent activating mutations include E709K, G719X, S768I, L861Q, and short in-frame insertion mutations in exon 20[9,10,11,12]. EGFR activating mutations in lung cancer tend to be mutually exclusive to KRAS activating mutations[13]. In contrast, a different set of recurrent activating EGFR mutations in the extracellular domain include R108K, A289V and G598V and are primarily observed in glioblastoma[8,14]. Amplification of EGFR is observed in several cancer types including 30% of glioblastoma, 12% of esophageal cancer, 10% of head and neck cancer, 5% of bladder cancer, and 5% of lung squamous cell carcinoma[5,6,7,14,15]. Deletion of exons 2-7, encoding the extracellular domain of EGFR (EGFRvIII), results in overexpression of a ligand-independent constitutively active protein and is observed in approximately 30% of glioblastoma[16,17,18].

Potential relevance: Approved first-generation EGFR tyrosine kinase inhibitors (TKIs) include erlotinib[19] (2004) and gefitinib[20] (2015), which block the activation of downstream signaling by reversible interaction with the ATP-binding site. Although initially approved for advanced lung cancer, the discovery that drug sensitivity was associated with exon 19 and exon 21 activating mutations allowed first-generation TKIs to become subsequently approved for front-line therapy in lung cancer tumors containing exon 19 or exon 21 activating mutations. Second-generation TKIs afatinib[21] (2013) and dacomitinib[22] (2018) bind EGFR and other ERBB/HER gene family members irreversibly and were subsequently approved. First- and second-generation TKIs afatinib, dacomitinib, erlotinib, and gefitinib are recommended for the treatment NSCLC harboring EGFR exon 19 insertions, exon 19 deletions, point mutations L861Q,

Biomarker Descriptions (continued)

L858R, S768I, and codon 719 mutations, whereas most EGFR exon 20 insertions, except p.A763_Y764insFQEA, confer resistance to the same therapies[23,24,25,26]. However, in 2021, the irreversible tyrosine kinase inhibitor, mobocertinib[27]was FDA approved for the treatment of NSCLC with EGFR exon 20 insertion mutations. Additionally, in 2022, the FDA granted breakthrough therapy designation to the irreversible EGFR inhibitors, CLN-081 (TPC-064)[28] and sunvozertinib[29], for locally advanced or metastatic non-small cell lung cancer harboring EGFR exon 20 insertion mutations. In lung cancer containing EGFR exon 19 or 21 activating mutations, treatment with TKIs is eventually associated with the emergence of drug resistance[30]. The primary resistance mutation that emerges following treatment with first-generation TKI is T790M, accounting for 50-60% of resistant cases[8]. Third generation TKIs were developed to maintain sensitivity in the presence of T790M. Osimertinib[31] (2015) is an irreversible inhibitor indicated for metastatic EGFR T790M positive lung cancer and for the first-line treatment of metastatic NSCLC containing EGFR exon 19 deletions or exon 21 L858R mutations. Like first-generation TKIs, treatment with osimertinib is associated with acquired resistance. In this case, resistance is associated with the C797S mutation and occurs in 22-44% of cases[30]. The T790M and C797S mutations may be each selected following sequential treatment with a first-generation TKI followed by a third-generation TKI or vice versa[32]. T790M and C797S can occur in either cis or trans allelic orientation[32]. If C797S is observed following progression after treatment with a third-generation TKI in the first-line setting, sensitivity may be retained to first-generation TKIs[32]. If C797S co-occurs in trans with T790M following sequential treatment with first- and third-generation TKIs, patients may exhibit sensitivity to combination first- and third-generation TKIs, but resistance to third-generation TKIs alone[32,33]. However, C797S occurring in cis conformation with T790M, confers resistance to first- and third-generation TKIs[32]. Fourth-generation TKIs are in development to overcome acquired C797S and T790M resistance mutations after osimertinib treatment. EGFR targeting antibodies including cetuximab (2004), panitumumab (2006), and necitumumab (2016) are under investigation in combination with EGFR-targeting TKIs for efficacy against EGFR mutations. The bispecific antibody, amivantamab[34], targeting EGFR and MET was approved (2021) NSCLC tumors harboring EGFR exon 20 insertion mutations. The Oncoprex immunogene therapy quaratusugene ozeplasmid[35] in combination with osimertinib received a fast track designation from the FDA (2020) for NSCLC tumors harboring EGFR mutations that progressed on osimertinib alone. BDTX-189[36] was granted a fast track designation (2020) for the treatment of solid tumors harboring an EGFR exon 20 insertion mutation.

ERBB3 (erb-b2 receptor tyrosine kinase 3)

Background: The ERBB3 gene encodes the erb-b2 receptor tyrosine kinase 3, a member of the human epidermal growth factor receptor (HER) family. Along with ERBB3/HER3, EGFR/ERBB1/HER1, ERBB2/HER2, and ERBB4/HER4 make up the HER protein family[1]. ERBB3/HER3 binds to extracellular factors, such as neuregulins, but has an impaired kinase domain[37]. Upon ligand binding, ERBB3 forms hetero-dimers with other ERBB/HER family members, including ERBB2/HER2 resulting in activation of tyrosine kinase activity primarily through its dimerization partner.

Alterations and prevalence: ERBB3 gene amplification leading to an increase in expression occurs at low frequency (1-5%) in several cancer types including bladder, esophagus, lung adenocarcinoma, ovarian, pancreas, sarcoma, stomach, and uterine cancers[5,6,7,38,39,40,41]. ERBB3 is also the target of relatively frequent (5-10%) and recurrent somatic mutations in diverse cancer types including bladder, cervical, colorectal, and stomach cancers[6,7,38,40,42]. Recurrent ERBB3 mutations such as V104L/M, occur primarily in the extracellular domain.

Potential relevance: Currently, no therapies are approved for ERBB3 aberrations. Overexpression and activation of ERBB3/HER3 is one mechanism of acquired resistance to therapies targeting EGFR and ERBB2/HER2[43,44]. Preclinical and translational research studies have characterized the oncogenic potential of recurrent ERBB3 mutations and their sensitivity to anti-ERBB antibodies and small molecule inhibitors[45,46,47,48]. A phase I study exhibited progression-free survival (PFS) of 2.5 months and overall survival (OS) of 9 months in 25 patients with ERBB3 mutations treated by anti-ERBB antibodies or molecular-targeted agents[49].

JAK3 (Janus kinase 3)

Background: The JAK3 gene encodes a non-receptor, membrane associated protein tyrosine kinase (PTK). JAK3 is a member of the Janus kinase (JAK) family that includes JAK1, JAK2, JAK3, and TYK2. Janus kinases are characterized by the presence of a second phosphotransferase-related or pseudokinase domain immediately N-terminal to the PTK domain[50]. JAK kinases function with signal transducer and activator of transcription (STAT) proteins to facilitate intracellular signal transduction required for cytokine receptor and interferon-alpha/beta/gamma signaling[50,51,52].

Alterations and prevalence: Recurrent somatic mutations in JAK3 have been observed in T-cell lymphomas and acute lymphoblastic leukemia (ALL)[53,54]. Mutations in the pseudokinase domain (M511I, A573V, R657W), and kinase domain (L857Q) activate the JAK/STAT pathway and transform hematopoietic cells in vitro[53]. These variants are infrequently observed in solid cancers[6].

Potential relevance: Currently, no therapies are approved for JAK3 aberrations. Tofacitinib (2012) is a JAK3 inhibitor FDA approved for rheumatoid and psoriatic arthritis. Activating mutations in JAK3, including the germline variant V722I, promoted increased expression of PD-L1 in lung cancer and were associated with durable benefit from tofacitinib PD-L1 blockade[55].

Biomarker Descriptions (continued)

MYC (MYC proto-oncogene, bHLH transcription factor)

Background: The MYC gene encodes the MYC proto-oncogene (c-MYC), a basic helix-loop-helix transcription factor that regulates the expression of numerous genes that control cell cycle progression, apoptosis, metabolic pathways, and cellular transformation[56,57,58,59]. MYC is part of the MYC oncogene family that includes related transcription factors MYCN and MYCL that regulate transcription in 10-15% of promoter regions[60]. MYC functions as a heterodimer in complex with the transcription factor MAX[57,61].

Alterations and prevalence: Recurrent somatic alterations are observed in both solid and hematological cancers. Recurrent somatic mutations in MYC, including codon T58, are infrequent and hypothesized to increase the stability of the MYC protein[62,63]. MYC gene amplification is particularly common in diverse solid tumors. MYC amplification is observed in 30% of serous ovarian cancer, 20% of uterine serous carcinoma, 15% of esophageal and breast cancers, and is common (1-10%) in numerous other cancer types[7,64,65]. MYC is the target of the t(8;14)(q24;32) chromosomal translocation in Burkitt's lymphoma that places MYC coding sequences adjacent to immunoglobulin region regulatory sequences, which results in increased MYC expression[66,67].

Potential relevance: Currently, no therapies are approved for MYC aberrations. Due to the high frequency of somatic MYC alterations in cancer, many approaches are being investigated in clinical trials including strategies to disrupt complex formation with MAX, including inhibition of MYC expression and synthetic lethality associated with MYC overexpression[56,68,69,70].

Relevant Therapy Summary

● In this cancer type ○ In other cancer type ◑ In this cancer type and other cancer types ✖ No evidence

EGFR exon 20 insertion

Relevant Therapy	FDA	NCCN	EMA	ESMO	Clinical Trials*
amivantamab	●	●	●	✖	● (I)
mobocertinib	●	●	✖	✖	● (III)
amivantamab, chemotherapy	✖	✖	✖	✖	● (III)
sunvozertinib	✖	✖	✖	✖	● (I/II)
TPC-064	✖	✖	✖	✖	● (I/II)
lazertinib, amivantamab	✖	✖	✖	✖	● (I)
TNO-155, nazartinib	✖	✖	✖	✖	● (I)

EGFR c.2284-1G>C

Relevant Therapy	FDA	NCCN	EMA	ESMO	Clinical Trials*
sunvozertinib	✖	✖	✖	✖	● (I/II)
amivantamab	✖	✖	✖	✖	● (I)
lazertinib, amivantamab, chemotherapy	✖	✖	✖	✖	● (I)

* Most advanced phase (IV, III, II/III, II, I/II, I) is shown and multiple clinical trials may be available.

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Relevant Therapy Details

Current FDA Information

● In this cancer type ○ In other cancer type ◑ In this cancer type and other cancer types

FDA information is current as of 2022-08-17. For the most up-to-date information, search www.fda.gov.

EGFR exon 20 insertion

amivantamab

Cancer type: Non-Small Cell Lung Cancer **Label as of:** 2021-12-21 **Variant class:** EGFR exon 20 insertion

Indications and usage:

RYBREVANT® is a bispecific EGF receptor-directed and MET receptor directed antibody indicated for the treatment of adult patients with locally advanced or metastatic non-small cell lung cancer (NSCLC) with epidermal growth factor receptor (EGFR) exon 20 insertion mutations, as detected by an FDA-approved test, whose disease has progressed on or after platinum-based chemotherapy.

This indication is approved under accelerated approval based on overall response rate and duration of response. Continued approval for this indication may be contingent upon verification and description of clinical benefit in the confirmatory trials.

Reference:

https://www.accessdata.fda.gov/drugsatfda_docs/label/2021/761210s001lbl.pdf

mobocertinib

Cancer type: Non-Small Cell Lung Cancer **Label as of:** 2021-09-15 **Variant class:** EGFR exon 20 insertion

Indications and usage:

EXKIVITY™ is a kinase inhibitor indicated for the treatment of adult patients with locally advanced or metastatic non-small cell lung cancer (NSCLC) with epidermal growth factor receptor (EGFR) exon 20 insertion mutations, as detected by an FDA-approved test, whose disease has progressed on or after platinum-based chemotherapy.

This indication is approved under accelerated approval based on overall response rate and duration of response. Continued approval for this indication may be contingent upon verification and description of clinical benefit in confirmatory trial(s).

Reference:

https://www.accessdata.fda.gov/drugsatfda_docs/label/2021/215310s000lbl.pdf

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Current NCCN Information

● In this cancer type ○ In other cancer type ◑ In this cancer type and other cancer types

NCCN information is current as of 2022-08-01. For the most up-to-date information, search www.nccn.org.
For NCCN International Adaptations & Translations, search www.nccn.org/global/international_adaptations.aspx.

EGFR exon 20 insertion

amivantamab

Cancer type: Non-Small Cell Lung Cancer **Variant class:** EGFR exon 20 insertion

NCCN Recommendation category: 2A

Population segment (Line of therapy):
- Adenocarcinoma, Large Cell, Squamous Cell, Not otherwise specified (NOS); Advanced, Metastatic, Progression (Subsequent therapy)

Reference: NCCN Guidelines® - NCCN-Non-Small Cell Lung Cancer [Version 3.2022]

mobocertinib

Cancer type: Non-Small Cell Lung Cancer **Variant class:** EGFR exon 20 insertion

NCCN Recommendation category: 2A

Population segment (Line of therapy):
- Adenocarcinoma, Large Cell, Squamous Cell, Not otherwise specified (NOS); Advanced, Metastatic, Progression (Subsequent therapy)

Reference: NCCN Guidelines® - NCCN-Non-Small Cell Lung Cancer [Version 3.2022]

Disclaimer: The data presented here is from a curated knowledgebase of publicly available information, but may not be exhaustive. The data version is 2022.09(005).

Current EMA Information

● In this cancer type ○ In other cancer type ◑ In this cancer type and other cancer types

EMA information is current as of 2022-08-17. For the most up-to-date information, search www.ema.europa.eu/ema.

EGFR exon 20 insertion

● amivantamab

Cancer type: Non-Small Cell Lung Cancer **Label as of:** 2022-02-01 **Variant class:** EGFR exon 20 insertion

Reference:

https://www.ema.europa.eu/en/documents/product-information/rybrevant-epar-product-information_en.pdf

Disclaimer: The data presented here is from a curated knowledgebase of publicly available information, but may not be exhaustive. The data version is 2022.09(005).

Clinical Trials in Taiwan region:

Clinical Trials Summary

EGFR exon 20 insertion

NCT ID	Title	Phase
NCT04538664	A Randomized, Open-label Phase III Study of Combination Amivantamab and Carboplatin-Pemetrexed Therapy, Compared With Carboplatin-Pemetrexed, in Patients With EGFR Exon 20ins Mutated Locally Advanced or Metastatic Non-Small Cell Lung Cancer	III
NCT04129502	A Randomized Phase III Multicenter Open-label Study to Compare the Efficacy of TAK-788 as First-line Treatment Versus Platinum-Based Chemotherapy in Patients With Non-Small Cell Lung Cancer With EGFR Exon 20 Insertion Mutations	III
NCT03974022	A Phase I/II, Open-Label, Multicenter Study to Assess the Safety, Tolerability, Pharmacokinetics and Anti-tumor Efficacy of DZD9008 in Patients With Advanced Non-Small Cell Lung Cancer (NSCLC) with EGFR or HER2 Mutation	I/II
NCT04036682	A Phase I/IIa, Open-Label, Multi-Center Trial To Assess Safety, Tolerability, Pharmacokinetics, Pharmacodynamics, And Efficacy Of CLN-081 In Patients With Non-Small Cell Lung Cancer Harboring EGFR Exon 20 Insertion Mutations	I/II
NCT02609776	A Phase I, First-in-Human, Open-Label, Dose Escalation Study of JNJ-61186372, a Human Bispecific EGFR and cMet Antibody, in Subjects With Advanced Non-Small Cell Lung Cancer.	I
NCT04077463	An Open-label Phase I/Ib Study to Evaluate the Safety and Pharmacokinetics of JNJ-73841937 (Lazertinib), a Third Generation EGFR-TKI, as Monotherapy or in Combinations With JNJ-61186372, a Human Bispecific EGFR and cMet Antibody in Participants With Advanced Non-Small Cell Lung Cancer	I
NCT03114319	An Open-label, Multi-center, Phase I, Dose Finding Study of Oral TNO155 in Adult Patients With Advanced Solid Tumors.	I

EGFR c.2284-1G>C

NCT ID	Title	Phase
NCT02609776	A Phase I, First-in-Human, Open-Label, Dose Escalation Study of JNJ-61186372, a Human Bispecific EGFR and cMet Antibody, in Subjects With Advanced Non-Small Cell Lung Cancer.	I
NCT03974022	A Phase I/II, Open-Label, Multicenter Study to Assess the Safety, Tolerability, Pharmacokinetics and Anti-tumor Efficacy of DZD9008 in Patients With Advanced Non-Small Cell Lung Cancer (NSCLC) with EGFR or HER2 Mutation	I/II
NCT04077463	An Open-label Phase I/Ib Study to Evaluate the Safety and Pharmacokinetics of JNJ-73841937 (Lazertinib), a Third Generation EGFR-TKI, as Monotherapy or in Combinations With JNJ-61186372, a Human Bispecific EGFR and cMet Antibody in Participants With Advanced Non-Small Cell Lung Cancer	I

Disclaimer: The data presented here is from a curated knowledgebase of publicly available information, but may not be exhaustive. The data version is 2022.09(005).

Alerts Informed By Public Data Sources

Current FDA Information

Contraindicated Not recommended Resistance Breakthrough Fast Track

FDA information is current as of 2022-08-17. For the most up-to-date information, search www.fda.gov.

EGFR exon 20 insertion

sunvozertinib

Cancer type: Non-Small Cell Lung Cancer **Variant class:** EGFR exon 20 insertion

Supporting Statement:
The FDA has granted Breakthrough Therapy Designation to a selective, irreversible, novel epidermal growth factor receptor (EGFR) inhibitor, DZD9008 (Sunvozertinib) for EGFR exon 20 insertion mutation positive locally advanced or metastatic non-small cell lung cancer.

Reference:

https://www.biospace.com/article/releases/fda-grants-breakthrough-therapy-designation-for-dizal-pharmaceutical-s-dzd9008-in-patients-with-locally-advanced-or-metastatic-non-small-cell-lung-cancer-harboring-egfr-exon20-insertion/

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sunvozertinib

Cancer type: Non-Small Cell Lung Cancer **Variant class:** EGFR exon 20 insertion

Supporting Statement:
The FDA has granted Breakthrough Designation to a small molecule inhibitor, DZD9008 (sunvozertinib), for the treatment of patients with locally advanced or metastatic non-small cell lung cancer (NSCLC) with epidermal growth factor receptor (EGFR) exon20 insertion mutations whose disease has progressed on or after platinum-based chemotherapy.

Reference:

https://www.prnewswire.com/news-releases/fda-grants-breakthrough-therapy-designation-for-dizal-pharmaceuticals-dzd9008-in-patients-with-locally-advanced-or-metastatic-non-small-cell-lung-cancer-harboring-egfr-exon20-insertion-301469692.html

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TPC-064

Cancer type: Non-Small Cell Lung Cancer **Variant class:** EGFR exon 20 insertion

Supporting Statement:
The FDA has granted Breakthrough Therapy Designation to an irreversible EGFR inhibitor, CLN-081, for locally advanced or metastatic non-small cell lung cancer harboring EGFR exon 20 insertion mutations who have previously received platinum-based systemic chemotherapy.

Reference:

https://investors.cullinanoncology.com/news-releases/news-release-details/fda-grants-breakthrough-therapy-designation-cullinan-oncologys

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EGFR exon 20 insertion (continued)

BDTX-189

Cancer type: Solid Tumor **Variant class:** EGFR exon 20 insertion

Supporting Statement:
The FDA has granted Fast Track Designation to BDTX-189 for solid tumors harboring a HER2 mutation or an EGFR or HER2 exon 20 insertion after progression on prior therapy.

Reference:

https://investors.blackdiamondtherapeutics.com/news-releases/news-release-details/black-diamond-therapeutics-granted-fast-track-designation-fda

osimertinib + quaratusugene ozeplasmid

Cancer type: Non-Small Cell Lung Cancer **Variant class:** EGFR mutation

Supporting Statement:
The FDA has granted Fast Track Designation to the immunogene therapy, quaratusugene ozeplasmid, in combination with EGFR inhibitor osimertinib for the treatment of non-small cell lung cancer (NSCLC) with EFGR mutations that progressed after treatment with osimertinib alone.

Reference:

https://www.genprex.com/news/genprex-receives-u-s-fda-fast-track-designation-for-gene-therapy-that-targets-lung-cancer/

EGFR c.2284-1G>C

osimertinib + quaratusugene ozeplasmid

Cancer type: Non-Small Cell Lung Cancer **Variant class:** EGFR mutation

Supporting Statement:
The FDA has granted Fast Track Designation to the immunogene therapy, quaratusugene ozeplasmid, in combination with EGFR inhibitor osimertinib for the treatment of non-small cell lung cancer (NSCLC) with EFGR mutations that progressed after treatment with osimertinib alone.

Reference:

https://www.genprex.com/news/genprex-receives-u-s-fda-fast-track-designation-for-gene-therapy-that-targets-lung-cancer/

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Current NCCN Information

Contraindicated Not recommended Resistance Breakthrough Fast Track

NCCN information is current as of 2022-08-01. For the most up-to-date information, search www.nccn.org.
For NCCN International Adaptations & Translations, search www.nccn.org/global/international_adaptations.aspx.

EGFR exon 20 insertion

atezolizumab

Cancer type: Non-Small Cell Lung Cancer **Variant class:** EGFR mutation

Summary:

NCCN Guidelines® include the following supporting statement(s):

- "subsequent therapy with pembrolizumab, nivolumab, or atezolizumab is not recommended in patients with EGFR mutations or ALK rearrangements."

Reference: NCCN Guidelines® - NCCN-Non-Small Cell Lung Cancer [Version 3.2022]

nivolumab

Cancer type: Non-Small Cell Lung Cancer **Variant class:** EGFR mutation

Summary:

NCCN Guidelines® include the following supporting statement(s):

- "subsequent therapy with pembrolizumab, nivolumab, or atezolizumab is not recommended in patients with EGFR mutations or ALK rearrangements."

Reference: NCCN Guidelines® - NCCN-Non-Small Cell Lung Cancer [Version 3.2022]

pembrolizumab

Cancer type: Non-Small Cell Lung Cancer **Variant class:** EGFR mutation

Summary:

NCCN Guidelines® include the following supporting statement(s):

- "subsequent therapy with pembrolizumab, nivolumab, or atezolizumab is not recommended in patients with EGFR mutations or ALK rearrangements."

Reference: NCCN Guidelines® - NCCN-Non-Small Cell Lung Cancer [Version 3.2022]

afatinib

Cancer type: Non-Small Cell Lung Cancer **Variant class:** EGFR exon 20 insertion

Summary:
NCCN Guidelines® include the following supporting statement(s):

- "Patients with EGFR exon 20 insertion mutations are usually resistant to erlotinib, gefitinib, afatinib, or dacomitinib, although there are rare exceptions."

Reference: NCCN Guidelines® - NCCN-Non-Small Cell Lung Cancer [Version 3.2022]

Disclaimer: The data presented here is from a curated knowledgebase of publicly available information, but may not be exhaustive. The data version is 2022.09(005).

EGFR exon 20 insertion (continued)

dacomitinib

Cancer type: Non-Small Cell Lung Cancer **Variant class:** EGFR exon 20 insertion

Summary:
NCCN Guidelines® include the following supporting statement(s):

- "Patients with EGFR exon 20 insertion mutations are usually resistant to erlotinib, gefitinib, afatinib, or dacomitinib, although there are rare exceptions."

Reference: NCCN Guidelines® - NCCN-Non-Small Cell Lung Cancer [Version 3.2022]

erlotinib

Cancer type: Non-Small Cell Lung Cancer **Variant class:** EGFR exon 20 insertion

Summary:
NCCN Guidelines® include the following supporting statement(s):

- "Patients with EGFR exon 20 insertion mutations are usually resistant to erlotinib, gefitinib, afatinib, or dacomitinib, although there are rare exceptions."

Reference: NCCN Guidelines® - NCCN-Non-Small Cell Lung Cancer [Version 3.2022]

gefitinib

Cancer type: Non-Small Cell Lung Cancer **Variant class:** EGFR exon 20 insertion

Summary:
NCCN Guidelines® include the following supporting statement(s):

- "Patients with EGFR exon 20 insertion mutations are usually resistant to erlotinib, gefitinib, afatinib, or dacomitinib, although there are rare exceptions."

Reference: NCCN Guidelines® - NCCN-Non-Small Cell Lung Cancer [Version 3.2022]

EGFR c.2284-1G>C

atezolizumab

Cancer type: Non-Small Cell Lung Cancer **Variant class:** EGFR mutation

Summary:

NCCN Guidelines® include the following supporting statement(s):

- "subsequent therapy with pembrolizumab, nivolumab, or atezolizumab is not recommended in patients with EGFR mutations or ALK rearrangements."

Reference: NCCN Guidelines® - NCCN-Non-Small Cell Lung Cancer [Version 3.2022]

Disclaimer: The data presented here is from a curated knowledgebase of publicly available information, but may not be exhaustive. The data version is 2022.09(005).

EGFR c.2284-1G>C (continued)

nivolumab

Cancer type: Non-Small Cell Lung Cancer **Variant class:** EGFR mutation

Summary:

NCCN Guidelines® include the following supporting statement(s):

- "subsequent therapy with pembrolizumab, nivolumab, or atezolizumab is not recommended in patients with EGFR mutations or ALK rearrangements."

Reference: NCCN Guidelines® - NCCN-Non-Small Cell Lung Cancer [Version 3.2022]

pembrolizumab

Cancer type: Non-Small Cell Lung Cancer **Variant class:** EGFR mutation

Summary:

NCCN Guidelines® include the following supporting statement(s):

- "subsequent therapy with pembrolizumab, nivolumab, or atezolizumab is not recommended in patients with EGFR mutations or ALK rearrangements."

Reference: NCCN Guidelines® - NCCN-Non-Small Cell Lung Cancer [Version 3.2022]

Current EMA Information

Contraindicated Not recommended Resistance Breakthrough Fast Track

EMA information is current as of 2022-08-17. For the most up-to-date information, search www.ema.europa.eu/ema.

EGFR exon 20 insertion

gefitinib

Cancer type: Non-Small Cell Lung Cancer **Label as of:** 2022-07-05 **Variant class:** EGFR exon 20 insertion

Reference:

https://www.ema.europa.eu/en/documents/product-information/iressa-epar-product-information_en.pdf

gefitinib (Mylan)

Cancer type: Non-Small Cell Lung Cancer **Label as of:** 2022-06-16 **Variant class:** EGFR exon 20 insertion

Reference:

https://www.ema.europa.eu/en/documents/product-information/gefitinib-mylan-epar-product-information_en.pdf

Disclaimer: The data presented here is from a curated knowledgebase of publicly available information, but may not be exhaustive. The data version is 2022.09(005).

Signatures

Testing Personnel: Shu-Ying Wang 2022.10.28 16:54:45 +08'00'

Laboratory Supervisor: 2022.10.28 16:54:57 +08'00'

Pathologist: Yi-Chu Yeh 2022.10.28 16:55:11 +08'00'

Disclaimer: The data presented here is from a curated knowledgebase of publicly available information, but may not be exhaustive. The data version is 2022.09(005).

References

1. King et al. Amplification of a novel v-erbB-related gene in a human mammary carcinoma. Science. 1985 Sep 6;229(4717):974-6. PMID: 2992089
2. Zhixiang. ErbB Receptors and Cancer. Methods Mol. Biol. 2017;1652:3-35. PMID: 28791631
3. Gutierrez et al. HER2: biology, detection, and clinical implications. Arch. Pathol. Lab. Med. 2011 Jan;135(1):55-62. PMID: 21204711
4. Pines et al. Oncogenic mutant forms of EGFR: lessons in signal transduction and targets for cancer therapy. FEBS Lett. 2010 Jun 18;584(12):2699-706. PMID: 20388509
5. Cancer Genome Atlas Research Network. Comprehensive molecular profiling of lung adenocarcinoma. Nature. 2014 Jul 31;511(7511):543-50. doi: 10.1038/nature13385. Epub 2014 Jul 9. PMID: 25079552
6. Weinstein et al. The Cancer Genome Atlas Pan-Cancer analysis project. Nat. Genet. 2013 Oct;45(10):1113-20. PMID: 24071849
7. Cerami et al. The cBio cancer genomics portal: an open platform for exploring multidimensional cancer genomics data. Cancer Discov. 2012 May;2(5):401-4. PMID: 22588877
8. da et al. EGFR mutations and lung cancer. Annu Rev Pathol. 2011;6:49-69. doi: 10.1146/annurev-pathol-011110-130206. PMID: 20887192
9. Arcila et al. EGFR exon 20 insertion mutations in lung adenocarcinomas: prevalence, molecular heterogeneity, and clinicopathologic characteristics. Mol. Cancer Ther. 2013 Feb;12(2):220-9. PMID: 23371856
10. Kobayashi et al. EGFR Exon 18 Mutations in Lung Cancer: Molecular Predictors of Augmented Sensitivity to Afatinib or Neratinib as Compared with First- or Third-Generation TKIs. Clin Cancer Res. 2015 Dec 1;21(23):5305-13. doi: 10.1158/1078-0432.CCR-15-1046. Epub 2015 Jul 23. PMID: 26206867
11. Yasuda et al. Structural, biochemical, and clinical characterization of epidermal growth factor receptor (EGFR) exon 20 insertion mutations in lung cancer. Sci Transl Med. 2013 Dec 18;5(216):216ra177. PMID: 24353160
12. Chiu et al. Epidermal Growth Factor Receptor Tyrosine Kinase Inhibitor Treatment Response in Advanced Lung Adenocarcinomas with G719X/L861Q/S768I Mutations. J Thorac Oncol. 2015 May;10(5):793-9. PMID: 25668120
13. Karachaliou et al. KRAS mutations in lung cancer. Clin Lung Cancer. 2013 May;14(3):205-14. PMID: 23122493
14. Brennan et al. The somatic genomic landscape of glioblastoma. Cell. 2013 Oct 10;155(2):462-77. PMID: 24120142
15. Cancer Genome Atlas Network. Comprehensive genomic characterization of head and neck squamous cell carcinomas. Nature. 2015 Jan 29;517(7536):576-82. PMID: 25631445
16. Mitsudomi et al. Epidermal growth factor receptor in relation to tumor development: EGFR gene and cancer. FEBS J. 2010 Jan;277(2):301-8. PMID: 19922469
17. Gazdar. Activating and resistance mutations of EGFR in non-small-cell lung cancer: role in clinical response to EGFR tyrosine kinase inhibitors. Oncogene. 2009 Aug;28 Suppl 1:S24-31. PMID: 19680293
18. Gan et al. The EGFRvIII variant in glioblastoma multiforme. J Clin Neurosci. 2009 Jun;16(6):748-54. PMID: 19324552
19. https://www.accessdata.fda.gov/drugsatfda_docs/label/2016/021743s025lbl.pdf
20. https://www.accessdata.fda.gov/drugsatfda_docs/label/2021/206995s004lbl.pdf
21. https://www.accessdata.fda.gov/drugsatfda_docs/label/2022/201292s017lbl.pdf
22. https://www.accessdata.fda.gov/drugsatfda_docs/label/2020/211288s003lbl.pdf
23. NCCN Guidelines® - NCCN-Non-Small Cell Lung Cancer [Version 3.2022]
24. Naidoo et al. Epidermal growth factor receptor exon 20 insertions in advanced lung adenocarcinomas: Clinical outcomes and response to erlotinib. Cancer. 2015 Sep 15;121(18):3212-3220. PMID: 26096453
25. Vyse et al. Targeting EGFR exon 20 insertion mutations in non-small cell lung cancer. Signal Transduct Target Ther. 2019;4:5. PMID: 30854234
26. Yi et al. A comparison of epidermal growth factor receptor mutation testing methods in different tissue types in non-small cell lung cancer. Int J Mol Med. 2014 Aug;34(2):464-74. PMID: 24891042
27. https://www.accessdata.fda.gov/drugsatfda_docs/label/2021/215310s000lbl.pdf
28. https://investors.cullinanoncology.com/news-releases/news-release-details/fda-grants-breakthrough-therapy-designation-cullinan-oncologys
29. https://www.prnewswire.com/news-releases/fda-grants-breakthrough-therapy-designation-for-dizal-pharmaceuticals-dzd9008-in-patients-with-locally-advanced-or-metastatic-non-small-cell-lung-cancer-harboring-egfr-exon20-insertion-301469692.html
30. Madic et al. EGFR C797S, EGFR T790M and EGFR sensitizing mutations in non-small cell lung cancer revealed by six-color crystal digital PCR. Oncotarget. 2018 Dec 21;9(100):37393-37406. PMID: 30647840

Disclaimer: The data presented here is from a curated knowledgebase of publicly available information, but may not be exhaustive. The data version is 2022.09(005).

References (continued)

31. https://www.accessdata.fda.gov/drugsatfda_docs/label/2022/208065s025lbl.pdf
32. Niederst et al. The Allelic Context of the C797S Mutation Acquired upon Treatment with Third-Generation EGFR Inhibitors Impacts Sensitivity to Subsequent Treatment Strategies. Clin. Cancer Res. 2015 Sep 1;21(17):3924-33. PMID: 25964297
33. Wang et al. Lung Adenocarcinoma Harboring EGFR T790M and In Trans C797S Responds to Combination Therapy of First- and Third-Generation EGFR TKIs and Shifts Allelic Configuration at Resistance. J Thorac Oncol. 2017 Nov;12(11):1723-1727. PMID: 28662863
34. https://www.accessdata.fda.gov/drugsatfda_docs/label/2021/761210s001lbl.pdf
35. https://www.genprex.com/news/genprex-receives-u-s-fda-fast-track-designation-for-gene-therapy-that-targets-lung-cancer/
36. https://investors.blackdiamondtherapeutics.com/news-releases/news-release-details/black-diamond-therapeutics-granted-fast-track-designation-fda
37. Knighton et al. Structural features that specify tyrosine kinase activity deduced from homology modeling of the epidermal growth factor receptor. Proc. Natl. Acad. Sci. U.S.A. 1993 Jun 1;90(11):5001-5. PMID: 8389462
38. Cancer Genome Atlas Research Network. Comprehensive molecular characterization of urothelial bladder carcinoma. Nature. 2014 Mar 20;507(7492):315-22. doi: 10.1038/nature12965. Epub 2014 Jan 29. PMID: 24476821
39. Cancer Genome Atlas Research Network. Integrated genomic analyses of ovarian carcinoma. Nature. 2011 Jun 29;474(7353):609-15. PMID: 21720365
40. Cancer Genome Atlas Research Network. Comprehensive molecular characterization of gastric adenocarcinoma. Nature. 2014 Sep 11;513(7517):202-9. doi: 10.1038/nature13480. Epub 2014 Jul 23. PMID: 25079317
41. Cancer et al. Integrated genomic characterization of endometrial carcinoma. Nature. 2013 May 2;497(7447):67-73. PMID: 23636398
42. Donna et al. Comprehensive molecular characterization of human colon and rectal cancer. Nature. 2012 Jul 18;487(7407):330-7. PMID: 22810696
43. Mujoo et al. Regulation of ERBB3/HER3 signaling in cancer. Oncotarget. 2014 Nov 15;5(21):10222-36. PMID: 25400118
44. Gaborit et al. Emerging anti-cancer antibodies and combination therapies targeting HER3/ERBB3. Hum Vaccin Immunother. 2016 Mar 3;12(3):576-92. PMID: 26529100
45. Mishra et al. Genomic alterations of ERBB receptors in cancer: clinical implications. Oncotarget. 2017 Dec 26;8(69):114371-114392. PMID: 29371993
46. Jaiswal et al. Oncogenic ERBB3 mutations in human cancers. Cancer Cell. 2013 May 13;23(5):603-17. PMID: 23680147
47. Zhang et al. HER3/ErbB3, an emerging cancer therapeutic target. Acta Biochim. Biophys. Sin. (Shanghai). 2016 Jan;48(1):39-48. PMID: 26496898
48. Ross et al. Targeting HER2 in colorectal cancer: The landscape of amplification and short variant mutations in ERBB2 and ERBB3. Cancer. 2018 Apr 1;124(7):1358-1373. PMID: 29338072
49. Verlingue et al. Human epidermal receptor family inhibitors in patients with ERBB3 mutated cancers: Entering the back door. Eur. J. Cancer. 2018 Mar;92:1-10. PMID: 29413684
50. Babon et al. The molecular regulation of Janus kinase (JAK) activation. Biochem. J. 2014 Aug 15;462(1):1-13. PMID: 25057888
51. Müller et al. The protein tyrosine kinase JAK1 complements defects in interferon-alpha/beta and -gamma signal transduction. Nature. 1993 Nov 11;366(6451):129-35. PMID: 8232552
52. Ren et al. JAK1 truncating mutations in gynecologic cancer define new role of cancer-associated protein tyrosine kinase aberrations. Sci Rep. 2013 Oct 24;3:3042. PMID: 24154688
53. Degryse et al. JAK3 mutants transform hematopoietic cells through JAK1 activation, causing T-cell acute lymphoblastic leukemia in a mouse model. Blood. 2014 Nov 13;124(20):3092-100. PMID: 25193870
54. Song et al. Oncogenic activation of the STAT3 pathway drives PD-L1 expression in natural killer/T-cell lymphoma. Blood. 2018 Sep 13;132(11):1146-1158. PMID: 30054295
55. Van et al. Long-term Benefit of PD-L1 Blockade in Lung Cancer Associated with JAK3 Activation. Cancer Immunol Res. 2015 Aug;3(8):855-63. PMID: 26014096
56. Chen et al. Targeting oncogenic Myc as a strategy for cancer treatment. Signal Transduct Target Ther. 2018 Feb 23;3:5. doi: 10.1038/s41392-018-0008-7. eCollection 2018. PMID: 29527331
57. Dang. MYC on the path to cancer. Cell. 2012 Mar 30;149(1):22-35. PMID: 22464321
58. Dominguez-Sola et al. Non-transcriptional control of DNA replication by c-Myc. Nature. 2007 Jul 26;448(7152):445-51. PMID: 17597761

Disclaimer: The data presented here is from a curated knowledgebase of publicly available information, but may not be exhaustive. The data version is 2022.09(005).

References (continued)

59. Wahlström et al. Impact of MYC in regulation of tumor cell metabolism. Biochim. Biophys. Acta. 2015 May;1849(5):563-9. PMID: 25038584
60. Dang et al. The c-Myc target gene network. Semin. Cancer Biol. 2006 Aug;16(4):253-64. PMID: 16904903
61. Blackwood et al. Myc and Max function as a nucleoprotein complex. Curr. Opin. Genet. Dev. 1992 Apr;2(2):227-35. PMID: 1638116
62. Chakraborty et al. A common functional consequence of tumor-derived mutations within c-MYC. Oncogene. 2015 Apr 30;34(18):2406-9. PMID: 24998853
63. Xu-Monette et al. Clinical and Biologic Significance of MYC Genetic Mutations in De Novo Diffuse Large B-cell Lymphoma. Clin. Cancer Res. 2016 Jul 15;22(14):3593-605. PMID: 26927665
64. Kalkat et al. MYC Deregulation in Primary Human Cancers. Genes (Basel). 2017 May 25;8(6). PMID: 28587062
65. Beroukhim et al. The landscape of somatic copy-number alteration across human cancers. Nature. 2010 Feb 18;463(7283):899-905. doi: 10.1038/nature08822. PMID: 20164920
66. Taub et al. Translocation of the c-myc gene into the immunoglobulin heavy chain locus in human Burkitt lymphoma and murine plasmacytoma cells. Proc Natl Acad Sci U S A. 1982 Dec;79(24):7837-41. PMID: 6818551
67. Ott et al. Understanding MYC-driven aggressive B-cell lymphomas: pathogenesis and classification. Hematology Am Soc Hematol Educ Program. 2013;2013:575-83. PMID: 24319234
68. Posternak et al. Strategically targeting MYC in cancer. F1000Res. 2016;5. PMID: 27081479
69. Carabet et al. Therapeutic Inhibition of Myc in Cancer. Structural Bases and Computer-Aided Drug Discovery Approaches. Int J Mol Sci. 2018 Dec 29;20(1). PMID: 30597997
70. Shahbazi et al. The Bromodomain Inhibitor JQ1 and the Histone Deacetylase Inhibitor Panobinostat Synergistically Reduce N-Myc Expression and Induce Anticancer Effects. Clin. Cancer Res. 2016 May 15;22(10):2534-44. PMID: 26733615

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